



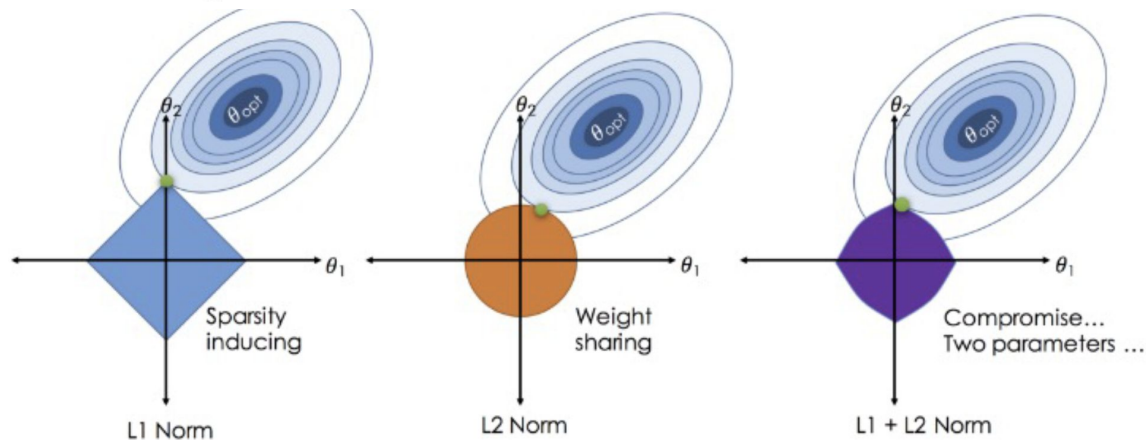
Elastic Net Regression

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What is Elastic Net Regression

- Combination of Ridge and Lasso Regression (a middle ground between both)

$$\hat{\beta} \equiv \underset{\beta}{\operatorname{argmin}} (\|y - X\beta\|^2 + \lambda_2 \|\beta\|^2 + \lambda_1 \|\beta\|_1).$$





Why use Elastic Net

- Handles multicollinearity very well, creating sparsity in coefficient and stabilizing coefficients
- More flexible than Ridge and Lasso
- Maintains the advantages of Ridge and Lasso Regression
 - Penalty terms for highly correlated features (Ridge)
 - Feature Selection (Lasso)
- Fix penalties of Ridge and Lasso
 - Handles high dimensional data with few examples far better than Lasso
 - Handles feature selection of important features far better than Ridge

Main Con: More Hyper Parameters

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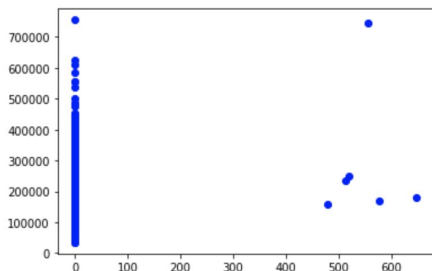
Dataset

Dataset: Housing Data

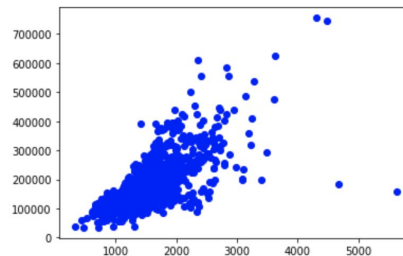
- Housing Data from Kaggle Competition
- Started with 36 numeric features
 - 17 Categorical Features useless to model (yearbuilt, yeargaragebuilt, etc)
- 19 Numerical Features used to predict Sales Price
 - Best features: Garage, 1st Floor Square Footage
 - Worst Features: Pool Area, Poarch Area



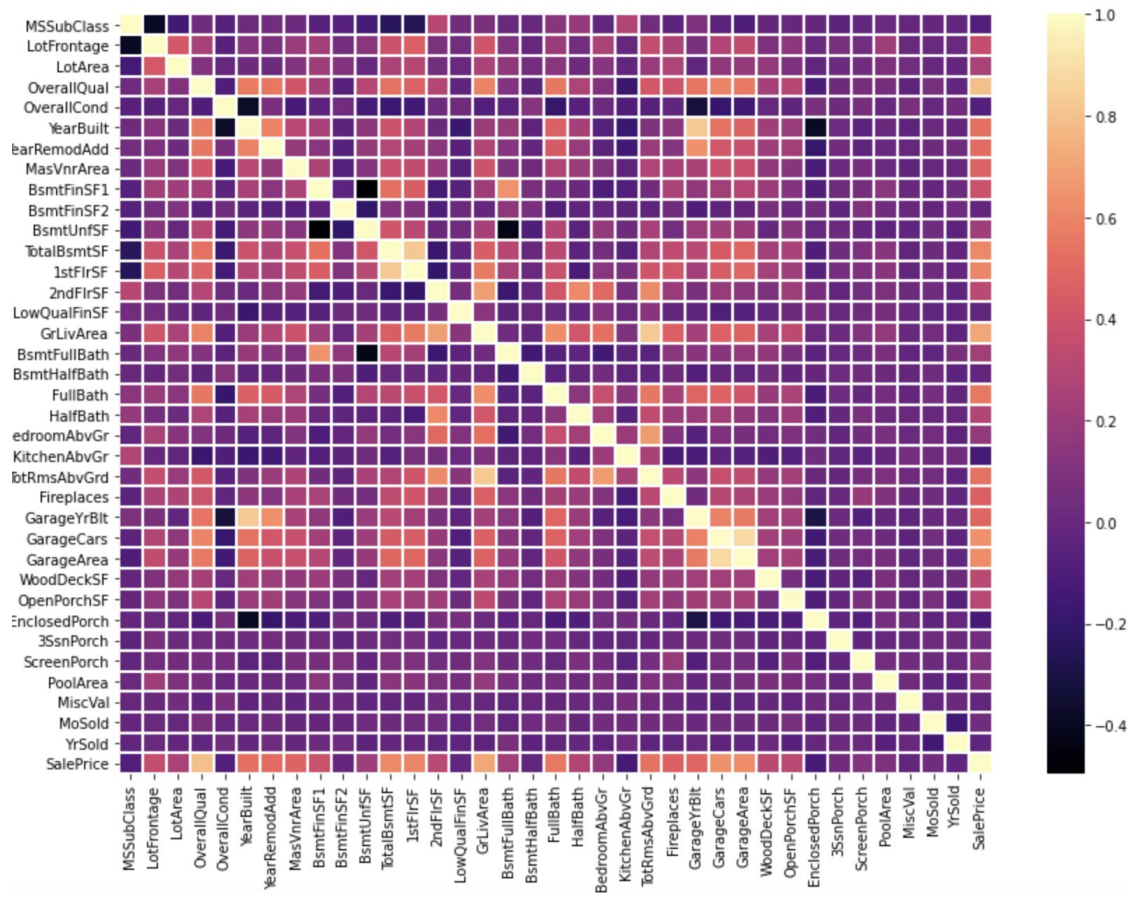
Pool Area:



General Living Area:



Dataset Correlation Matrix



Model Implementation



Elastic Net Implementation

- Alpha: Weight controlling implementation of Ridge and Lasso terms
- Mixing Parameter(λ): Ratio between Ridge and Lasso (1 = Lasso, 0 = Ridge)

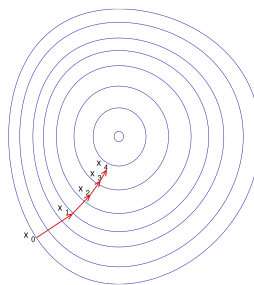
$$\sum_{i=1}^n \left(y_i - y_{(\text{pred})_i} \right)^2 + \alpha * \left(\lambda \sum_{j=1}^j |\theta_j| + (1-\lambda) \sum_{j=1}^j \theta_j^2 \right)$$

- Lambda_1 (Lasso): λ
- Lambda_2 (Ridge): $(1-\lambda)$
- Used same form as Sklearn to test numpy implementation against Sklearn's implementation

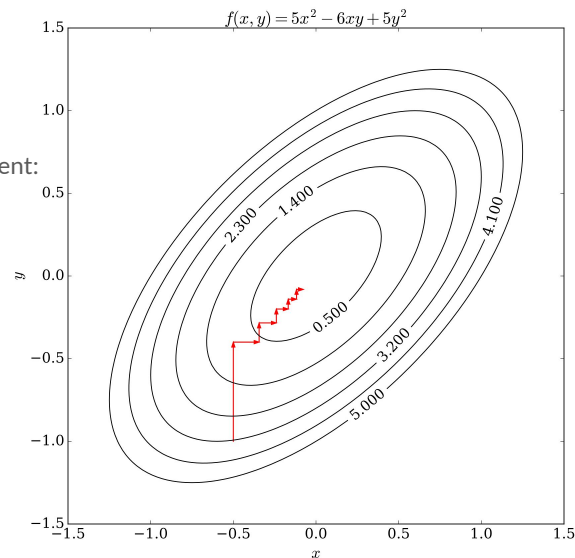
Optimization Technique

- Gradient Descent
 - Used as discussed in class
 - MSE term, L1 term, and L2 Term
 - Faster convergence
- Coordinate Descent
 - Loop through features and tune each parameter
 - Less computationally intense
 - Took significantly longer than gradient descent

Gradient Descent:

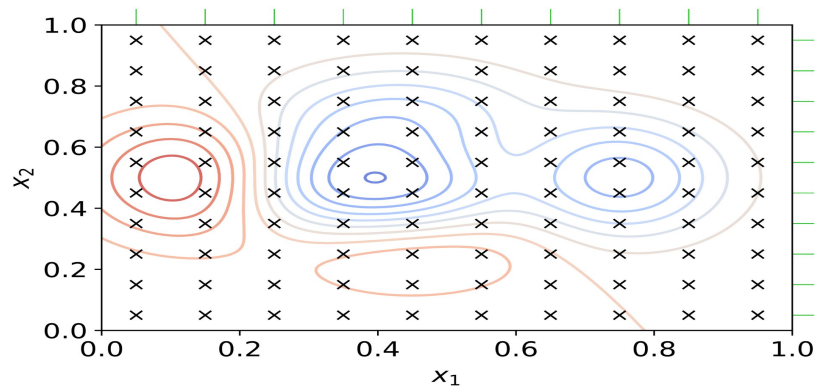


Coordinate Descent:



Hyper Parameter Tuning

- Rudimentary Grid Search
 - Simplistic method, systematic guess and check
 - Tested Alpha values from $[0, 1]$
 - Tested Mixing Parameter from $[0, .9]$





Results

- All three models converged to similar results, certain features were closer than others
- Coordinate Descent had better error than Gradient Descent by ~10%
- All models had an error of ~\$28000
 - Coordinate Descent, Sklearn, and Gradient Descent
- Optimal hyperparameters found via grid search
 - Optimal Alpha: 0.23076923076923075
 - Optimal Mixing Ratio: 0.8769230769230769
- Model was biased towards Lasso Regression



References

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC2929880/>
- https://scikit-learn.org/dev/modules/generated/sklearn.linear_model.ElasticNet.html
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